

CBSE Test Paper 01
Chapter 10 Mechanical Properties of Fluids

1. In a hydraulic lift the force applied on the smaller cylinder of area A_1 is F_1 . If the area of the larger cylinder is A_2 the maximum weight that can be lifted is **1**
 - a. F_1
 - b. $\frac{A_1}{A_2} F_1$
 - c. $F_1 A_2$
 - d. $\frac{A_2}{A_1} F_1$
2. A cylindrical jar of cross-sectional area 0.01 m^2 is filled with water to a height of 50 cm. It carries a tight-fitting piston of negligible mass. What is the pressure at the bottom of the jar when mass of 1 kg is placed on the piston? Take $g = 10 \text{ m s}^{-2}$ **1**
 - a. 6000 Pa
 - b. 4000 Pa
 - c. 2000 Pa
 - d. 5000 Pa
3. Pressure p at any point in a fluid at rest is **1**
 - a. the normal force at that point per unit volume
 - b. the force parallel to area at that point per unit area
 - c. the normal force at that point per unit mass
 - d. the normal force at that point per unit area
4. At large flow velocities the flow of a fluid becomes **1**
 - a. viscous
 - b. turbulent
 - c. compressible
 - d. laminar
5. According to Stokes' law if η is the coefficient of viscosity, 'R' the radius of the sphere

and it is moving with a velocity v the viscous drag force F is given by **1**

- a. $F_d = 6\pi\eta Rv$
- b. $F_d = 6\pi\eta R$
- c. $F_d = 2\pi\eta Rv$
- d. $F_d = 6\pi Rv$

- 6. What happens when a capillary tube of insufficient length is dipped in a liquid? **1**
 - 7. State the law of floatation? **1**
 - 8. If a wet piece of wood burns, then water droplets appear on the other end, why? **1**
 - 9. Find out the dimensions of coefficient of viscosity? **2**
 - 10. The surface tension and vapour pressure of water at 20°C is $7.28 \times 10^{-2} \text{ Nm}^{-1}$ and $2.33 \times 10^3 \text{ Pa}$, respectively. What is the radius of the smallest spherical water droplet which can form without evaporating at 20°C ? **2**
 - 11. Atmospheric pressure at a height of about 6 km decreases to nearly half of its value at the sea level, though the height of the atmosphere is more than 100 km. **2**
 - 12. Briefly explain Magnus effect. **3**
 - 13. A liquid drop of diameter 4mm breaks into 1000 droplets of equal size. Calculate the resultant change in the surface energy. Surface tension of the liquid is 0.07 N/m ? **3**
 - 14. A freshwater reservoir is 10 m deep. A horizontal pipe 2 cm in diameter passes through the reservoir 8.0 m below the water surface. A plug secures the pipe opening. At a certain time the plug is removed. What volume of water flows out of the pipe in 1 h? Assume cross-section area of the reservoir to be too large. **3**
 - 15. The surface tension of soap solution at 20°C is $2.50 \times 10^{-2} \text{ N m}^{-1}$. Calculate the excess pressure inside a soap bubble of radius 5 mm of this solution. If an air bubble of the same dimension were formed at depth of 40.0 cm inside a container containing the soap solution of relative density 1.20, what would be the pressure inside the bubble? ($1 \text{ atm} = 1.01 \times 10^5 \text{ pa.}$). **5**
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